ENSF 444 – Assignment 1

Schulich School of Engineering

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PART I: AI Ethics

1. **HireFast is now facing public criticism and potential regulatory scrutiny. Management must decide whether to modify, replace, or retire the system.**
    a. **Identify two sources of potential bias in the training data.**
        i. An automated hiring system relies heavily on its training data. If the said training data is based on seven years of historical hiring decisions, whether they were conscious or unconscious biases, then the model will learn and reproduce the same patterns. This issue has been observed in cases such as Amazon's hiring algorithm [1], where biased historical data led to discriminatory outcomes. As a result, certain groups may be indirectly disadvantaged by the system.
        ii. Additionally, if applicants from less prestigious universities or certain geographic regions were historically less likely to receive interviews, the model will learn to associate those backgrounds with lower hiring success. This creates a feedback loop in which underrepresented groups are consistently ranked lower. Similar concerns have been raised in research on automated hiring systems, where limited representation in training data leads to systematic down-ranking of specific groups [2].
    b. **Explain how proxy variables may cause discrimination even when protected attributes are excluded.**
        i. Even when protected attributes such as race or gender are explicitly removed. AI systems can still infer them through proxy variables. Studies have shown that models can effectively "rediscover" protected attributes due to structural inequalities that are embedded in the data itself, and there is no simple technical fix for this problem [4]. For example, features such as postal code, neighbourhood, or university attended often correlate strongly with socio-economic status, race, or gender. As a result, the model can indirectly reproduce discriminatory outcomes despite excluding of protected attributes.

2. **The system improves efficiency but may disadvantage certain groups.**
    a. **Discuss the ethical trade-offs between operational efficiency and fairness in this case.**
        i. HireFast's automated screening system significantly reduced time-to-hire and recruiter workload, clearly demonstrating gains in operational efficiency. However, this efficiency comes with ethical risks. Seattle University describes automated resume screening as "occupational segregation on speed," where existing biases are amplified across large applicant pods [2]. While the system processes applications faster, it

may systematically exclude qualified candidates from marginalized groups, raising serious concerns about fairness and equal opportunity.

b. **Who should be responsible for deciding what level of fairness is "acceptable" and why?**
    i. Regulators should play a key role by establishing baseline requirements for fairness and transparency. For example, Illinois' AI Video Interview Act mandates disclosure and consent when AI systems are used in hiring decisions [3]. However, legal compliance alone is not sufficient. Organizations themselves must define and update fairness standards that reflect their values and align with societal and applicant expectations. This responsibility typically falls to HR leadership, informed by input from stakeholders such as legal teams, ethicists, and affected employees. Research emphasizes that fair decisions involve moral judgement and social context and therefore can not be delegated solely to technical teams [4].

3. **Think about a time when you applied for a job, internship, scholarship, or program and were rejected or accepted without clear feedback.**
    a. **How did the lack (or presence) of feedback affect your perception of fairness?**
        i. A couple of months ago, I applied to Shell and even completed their pre-recorded interview. It took over a month to receive a response, and when I finally did, it was a generic rejection email with no explanation. While the rejection itself was disappointing, the lack of feedback was more frustrating. After being required to complete multiple stages, receiving no information about what went wrong made the whole process feel unfair. I couldn't tell whether I was rejected early, late, or because of something specific in my application. Without feedback, it's hard to know what to improve for next time, so the whole process felt a bit one-sided.
    b. **Would your opinion change if you knew the decision was made by an algorithm rather than a human?**
        i. If the message was the same generic rejection, I don't think it would change much. Whether it's a human or an algorithm, the result is still "no feedback, try again later." The only situation where it would matter is if a human took the time to add context or explain the decision. In that case, I'd prefer a human. But if both are sending the same template email, then it doesn't really make a difference to me.
    c. **Based on your experience, what level of transparency would you expect from an automated hiring system?**

i. I don't expect a full breakdown of the model or anything complicated, but I do think an automated system should at least tell me what part of my application was weak — whether it was experience, keywords, or something in the interview. Even a simple sentence would help. I don't need advice on how to fix it; just knowing what to fix would make the process feel more respectful. But I've also learned that automated systems rarely give meaningful feedback, so my expectations are low at this point.

4. **Propose two concrete changes (technical, organizational, or policy-based) that HireFest could implement to make the system more ethically responsible. For each change:**

Change 1: Limit proxy features and conduct fairness audits.
Change 2: Increase transparency and implement human-in-the-loop review.

a. **Explain how it addresses a specific ethical concern**
   i. Restricting or re-weighing features like postal code, neighbourhood, or university attended would help limit the influence of variables that act as proxies for protected attributes. In addition, running regular fairness audits would allow HireFast to check whether certain groups are being consistently ranked lower and adjust as issues are identified.
   ii. Being more transparent about the use of automated screening, along with conducting periodic audits, aligns with emerging regulatory expectations and helps maintain trust with applicants [3]. Adding human review prevents the algorithm from becoming the sole gatekeeper and reduces the risk of large-scale exclusion.
b. **Discuss one potential drawback**
   i. Removing strong predictors can reduce model accuracy or efficiency, reflecting the broader challenge that fairness improvements require performance trade-offs [4].
   ii. The drawback is increased operational cost and the need for additional training and oversight structures.

5. **Should automated resume screening systems be allowed in high-stakes hiring decisions?**
   a. **Argue for or against this position, using both technical and ethical reasoning.**
      i. Despite the flaws automated screening has, it also provides consistency and reduces some forms of human bias. In some cases, algorithms may even outperform human decision makers. This is highlighted in analyses of Amazon's system despite its flaws [1]. However, because

algorithmic bias can disproportionately harm marginalized groups and scale rapidly, high-stakes hiring decisions require human oversight. Research shows that without human influence on transparency and fairness safeguards, automated systems risk accelerating existing inequalities [2, 4].

PART II: Generative AI

1. **Should a company like HireFest Inc. be required to provide explanations to job applicants when an automated system rejects them? In your answer, please address:**
    a. **The trade-offs between transparency, model performance, and operational cost**
        i. Yes; Sure there are many factors to consider, but providing a reasonable level of transparency is a basic responsibility we have towards the candidates. I think an automated system should offer more information than a generic message like “we have chosen to move forward with another candidate.” At the same time, involving recruiters in reviewing both successful and rejected applications would significantly increase operational costs. Some compromise is necessary: the recruitment team must balance transparency with efficiency, and the model alone cannot be expected to generate detailed explanations without sacrificing performance.
    b. **Whether your answer would change depending on the role being hired for (e.g., internship vs. senior engineer)**
        i. Absolutely. The position and seniority of the role definitely influence the type and level of explanation that should be provided. For internships, there can be more leniency because students generally understand how many applications these roles receive, making detailed feedback difficult to provide. However, for more senior positions, this connects back to Part I, Q5: in high-stakes hiring decisions, there must be meaningful human oversight, transparency, and fairness safeguards.

2. **After writing your own answer, generate a response to the same question using a generative AI tool of your choice (e.g., ChatGPT, Claude, Gemini). Compare the two responses by addressing the following:**
    a. **One insight the AI response included that you did not**
        i. Gemini introduced specific legal and regulatory frameworks—such as the EU AI Act, Ontario’s 2026 AI disclosure laws, and the 4/5ths disparate-impact rule—as part of the justification for requiring explanations. My original answer focused on transparency, operational cost, and model performance, but you did not bring in external legal standards or future regulatory trends. This adds a broader policy context that strengthens the argument for mandatory explanations (Appendix B).

b. **One limitation, omission, or assumption in the AI response**

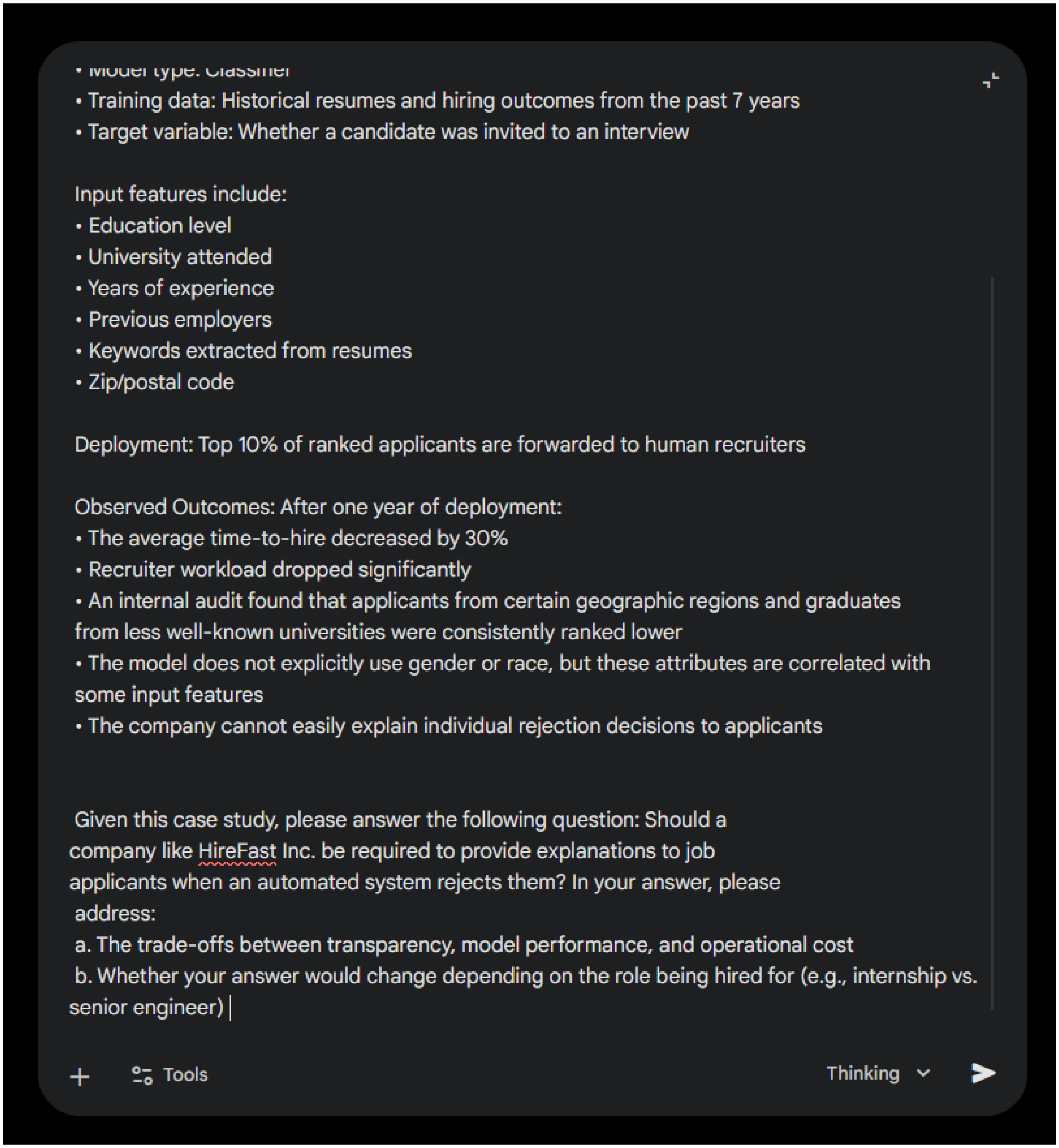
    i. Gemini assumes that detailed explanations are always feasible and beneficial, especially for internships, without acknowledging the operational realities I highlighted. For example, Gemini assumes that high-volume roles make individualized explanations extremely costly and sometimes impractical (Appendix B). To generalize, Gemini treats explanation as a straightforward solution, while overlooking the practical and organizational constraints involved.

c. **Whether the AI response changed your original opinion, and why or why not?**

    i. Gemini's answer didn't really change my overall opinion, but it did add some nuance. I still believe companies should provide at least a basic explanation when an automated system rejects someone, because it's a reasonable expectation of fairness. What Gemini's response did highlight, though, is how quickly regulations are evolving and how external legal pressure may eventually make explanations mandatory anyway. That context strengthened my argument but didn't shift my core view. My stance remains the same: transparency matters, but it has to be balanced with the practical limits of high-volume hiring systems.

Appendices

Appendix A: Prompt to Gemini



• Model type: Classifier
• Training data: Historical resumes and hiring outcomes from the past 7 years
• Target variable: Whether a candidate was invited to an interview

Input features include:
• Education level
• University attended
• Years of experience
• Previous employers
• Keywords extracted from resumes
• Zip/postal code

Deployment: Top 10% of ranked applicants are forwarded to human recruiters

Observed Outcomes: After one year of deployment:
• The average time-to-hire decreased by 30%
• Recruiter workload dropped significantly
• An internal audit found that applicants from certain geographic regions and graduates from less well-known universities were consistently ranked lower
• The model does not explicitly use gender or race, but these attributes are correlated with some input features
• The company cannot easily explain individual rejection decisions to applicants

Given this case study, please answer the following question: Should a company like HireFast Inc. be required to provide explanations to job applicants when an automated system rejects them? In your answer, please address:
a. The trade-offs between transparency, model performance, and operational cost
b. Whether your answer would change depending on the role being hired for (e.g., internship vs. senior engineer)

Appendix B: Gemini Response to Part II

Based on current legal trends (such as the **EU AI Act** and **Ontario's 2026 AI Disclosure laws**) and the specific risks identified in the HireFast audit, the company **should be required** to provide explanations. This requirement acts as a critical safeguard against "black box" discrimination.

a. Trade-offs: Transparency, Performance, and Cost

- **Model Performance vs. Interpretability:** Highly complex models (like Deep Learning) often yield better predictions but are "black boxes," whereas requiring explanations forces the use of more interpretable models which might have slightly lower raw accuracy.
- **Operational Cost vs. Litigation Risk:** While generating individual explanations for 50,000 applicants increases immediate computational and administrative costs, it significantly reduces the long-term risk of expensive class-action lawsuits regarding **disparate impact**.
- **Transparency vs. Gaming the System:** Providing detailed explanations could allow applicants to "keyword stuff" or manipulate their resumes to trick the algorithm, potentially degrading the quality of the candidate pool.
- **Data Quality vs. Bias Reinforcement:** Explanations expose when a model is using "proxy variables" (like Zip Code for race); without this transparency, HireFast effectively automates and scales historical prejudices under the guise of efficiency.

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b. Role Differentiation: Internships vs. Senior Engineers

- **Internships (High-Volume, Low-Signal):** Rejection explanations are **essential** here because candidates lack deep work history, making the model more likely to rely on biased proxies like "University Prestige" or "Zip Code."
- **Senior Engineers (Low-Volume, High-Signal):** Explanations are **easier to provide** but less legally "risky" since rejections are usually based on specific technical gaps (e.g., lacking a specific 10-year niche skill) rather than demographic proxies.
- **Entry-Level Vulnerability:** For early-career roles, an automated rejection without explanation can permanently discourage diverse talent from the industry, whereas senior candidates often have the professional network to understand why they were bypassed.
- **Regulatory Thresholds:** Many 2026 regulations (like Ontario's Bill 149) apply more strictly to high-volume "publicly advertised" roles, meaning the burden for transparency is legally higher for the internship-level roles HireFast is screening.

> **Legal Context (2026):** Under the **4/5ths Rule**, if HireFast's "Zip Code" filter results in a selection rate for one group that is less than 80% of another, they are legally liable for discrimination. Providing explanations is the only way to prove the decision was based on "business necessity" rather than bias.

Would you like me to draft a sample **Transparency Report** that HireFast could use to satisfy these 2026 regulatory requirements?

References

[1] Lavanchy, Maude. “Amazon’s Sexist Hiring Algorithm Could Still Be Better than a Human.” *IMD Business School for Management and Leadership Courses*, IMD, 10 Jan. 2025, www.imd.org/research-knowledge/digital/articles/amazons-sexist-hiring-algorithm-could-still-be-better-than-a-human/.

[2] Sedgwick, Geneva Lasprogata. “Facial Recognition in Hiring: Occupational Segregation on Speed.” *Seattle University*, 22 Apr. 2021, www.seattleu.edu/business/news-events/pov/ethics-matter/posts/facial-recognition-in-hiring-occupational-segregation-on-speed.php.

[3] “Ai Library Illinois.” *Morrison Foerster*, www.mofo.com/artificial-intelligence/illinois. Accessed 21 Jan. 2026.

[4] Kelan, Elisabeth. “No Quick Tech Fix for AI Bias against Women Job Applicants.” *University of Essex*, University of Essex, 3 Apr. 2024, www.essex.ac.uk/news/2024/04/03/no-quick-tech-fix-for-ai-bias-against-women-job-applicants.

Gen AI References & Notes

[5] "Part II Question 2" prompt. *Gemini*, 1.5 Flash version, Google, 29 Jan. 2026, https://gemini.google.com/app.

[6] “Grammatical and sentence fixes” prompt. *Microsoft Copilot*, Jan. 2026 version, Microsoft, 29 Jan. 2026, https://copilot.microsoft.com.